

Digest: Trivial Zeros of p -adic L -functions at Near Central Points

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1 Introduction

This document digests Denis Benois's paper on the trivial zeros of p -adic L -functions of modular forms. The paper proves a Mazur-Tate-Teitelbaum style formula for the derivative of p -adic L -functions at near central points.

2 The Main Theorem

Let f be a newform of weight k and character ε . For a p -adic L -function $L_{p,\alpha}(f, \omega^{k_0}, s)$ having a trivial zero at $s = k_0 \geq k/2$, the derivative at the near central point is given by:

$$L'_{p,\alpha}(f, \omega^{k_0}, k_0) = \ell_\alpha(f) \left(1 - \frac{\varepsilon(p)}{p}\right) \tilde{L}(f, k_0) \quad (1)$$

where $\ell_\alpha(f)$ is the generalized ℓ -invariant.

3 (φ, Γ) -modules and the ℓ -invariant

The construction relies heavily on the theory of (φ, Γ) -modules over the Robba ring $\mathcal{R}_L$. The ℓ -invariant $\ell(V, D)$ is defined via the Galois cohomology of the potentially semistable representation V and its crystalline submodule D .

4 Conclusion

This framework provides a unified proof of the trivial zero conjecture for elliptic modular forms, generalizing Greenberg's ℓ -invariant to non-ordinary cases.